The Periodic Table

Periods, groups, families and blocks

Name _____ Class _____ Date _____

YOU WILL NEED

A periodic table with groups numbered 1–18

Percent error = (predicted − actual) ÷ actual × 100 %

Show units in every calculation step.

A. Structure of the table

1. (a) What is the difference between a period and a group? (b) What determines the order of the elements?

2. Name the element and give its family (if it has one): (a) period 2, group 1 (b) period 4, group 17 (c) period 3, group 18 (d) period 4, group 2 (e) period 4, group 11

B. Metals, non-metals and metalloids

3. Classify each element as a metal, non-metal or metalloid: (a) potassium (b) phosphorus (c) germanium (d) copper (e) bromine. Then give two properties typical of metals.

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C. Valence electrons and ions

4. State the number of valence electrons and the usual ion formed by: (a) calcium (b) aluminium (c) oxygen (d) fluorine (e) nitrogen

5. Use the groups to predict the formulas of: (a) sodium oxide (b) calcium chloride (c) aluminium sulfide (d) magnesium nitride

D. Blocks

6. Which block is each element in? (a) Na (b) Fe (c) Br (d) U. Explain how the block is linked to the electrons.

E. History and patterns

7. Mendeleev predicted that eka-silicon would have an atomic mass of 72 and a density of 5.5 g/cm^3. Germanium has an atomic mass of 72.63 and a density of 5.32 g/cm^3. (a) Calculate the percent error of each prediction. (b) Why was leaving gaps in the table such a bold step?

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F. Challenge

8. Potassium reacts with water more vigorously than sodium, which reacts more vigorously than lithium. Fluorine is more reactive than chlorine. Suggest why reactivity increases down group 1 but decreases down group 17.

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